Question 1

A software company, TechNova Solutions, develops a mobile application that collects user data such as location, browsing habits, and personal preferences to improve its services. Recently, it was discovered that:

- The company shared user data with third-party advertisers without clear user consent.
- A hacker intercepted user data transmissions over an unsecured network.
- Some employees accessed confidential user information without authorization.
- Users have started receiving large amounts of unsolicited promotional emails after signing up for the app.

1) Using the scenario above, discuss the differences among morals, ethics, and law.
In your answer:

- Clearly define each term.
- Explain how each concept applies to the actions of TechNova Solutions.
- Provide one relevant example from the scenario for each term.
  (9 marks)

2)

Based on the scenario, discuss the following types of security attacks:

(i) Passive attacks

- Define passive attacks.
- Identify and explain one example from the scenario. (3 marks)

(ii) Active attacks

- Define active attacks.
- Identify and explain one example from the scenario. (3 marks)

3)

Users of the TechNova app are experiencing various types of spam after registration.

- List and describe FIVE (5) types of spam.
- For each type, relate it to how it could occur in the given scenario.
  (10 marks)

Question 2

Scenario:

A financial technology company, SecurePay Ltd, provides online payment services to millions of users. Due to rapid growth, the company recently faced several issues:

- A security audit revealed weak controls in managing sensitive customer data.
- Employees were unsure about proper security practices, leading to accidental data exposure.
- The company lacks a structured information security management system (ISMS).
- Management has decided to adopt international standards and improve its overall security posture.

a)

Referring to the scenario, management of SecurePay Ltd is considering implementing International Organization for Standardization standards.

- Briefly explain ISO/IEC 27001.
- List and describe THREE (3) categories (domains/clauses/controls) of ISO 27001 that would help SecurePay Ltd improve its security management.
- Relate your answer to the scenario where appropriate.
  (5 marks)

b)

To strengthen its system reliability and user confidence, SecurePay Ltd plans to adopt the four pillars of trustworthy computing.

- Briefly explain the FOUR (4) pillars of trustworthy computing.
- Illustrate how each pillar can help address the issues faced by SecurePay Ltd in the scenario.
  (8 marks)

c)

SecurePay Ltd intends to formalize its internal security practices.

- Discuss THREE (3) IT security policies that the company should adopt.
- For each policy:
    - Describe the policy.
    - Provide one example of how it would be applied in SecurePay Ltd based on the scenario.
      (12 marks)